

Concordia University
Department of Computer Science and Software
Engineering

SOEN 363: Data Systems for Software Engineers – Fall 2026

Project

(Phase 1)

Posted Date: September 19, 2026.

Due Date: October 20, 2025, at 11:59 PM.

Evaluation: 5% of final mark.

Late Submission: Please check the course outline.

Type: Group project.

Purpose: The purpose of this phase of the project is to design and implement a Database.

CEAB Attributes: This phase of the project is primarily evaluating your use of engineering tools.

1 Project Description

1.1 Overview

The purpose of this project is to design a database for a **Hospital Management System**. The system manages the complete journey of a patient, beginning with hospital registration and admission and continuing through triage, medical assessment, diagnostic testing, treatment, transfer between hospital units, diagnosis, and discharge.

A person receiving healthcare services from the hospital is identified as a **Patient**. Each patient is registered using a unique patient identifier that remains associated with the patient throughout their lifetime. The patient's record contains relatively stable information, including name, date of birth, gender, address, telephone number, emergency-contact information, and life status.

A patient may visit the hospital multiple times. Each hospital visit is recorded as a separate **Admission**. Therefore, one patient may have several admissions, but each admission belongs to exactly one patient.

Every admission records information such as:

- Admission date and time
- Admission type

- Reason for admission
- Admission source
- Insurance information
- Discharge date and time
- Final admission status

During each admission, the patient may receive one or more diagnoses. Each diagnosis must be represented using an **ICD-9 code**. Therefore, a patient does not necessarily have only one ICD-9 code. A patient may have several ICD-9 codes during one admission and may receive different ICD-9 codes during different admissions.

For example, a patient may receive ICD-9 code 401.9 for unspecified essential hypertension during one admission and another code for a cardiac or respiratory condition during a later admission.

1.2 What Is an ICD-9 Code?

ICD-9 stands for the **International Classification of Diseases, Ninth Revision**. It is the ninth revision of the international disease-classification system maintained and published by the **World Health Organization (WHO)**.

The WHO developed the International Classification of Diseases to provide a standardized method for recording, organizing, comparing, and reporting health information. ICD classifications group diseases and other health conditions into standardized categories and assign codes to those categories.

An ICD-9 code is therefore a standardized code representing a disease, injury, health condition, or cause of death. Instead of different hospitals using different names for the same condition, they can use the same internationally recognized code.

For example:

- ICD-9 code 401.9 represents *Unspecified essential hypertension*.
- ICD-9 code 250.00 represents *Diabetes mellitus without mention of complication, type II or unspecified type, not stated as uncontrolled*.
- ICD-9 code 410.9 represents *Acute myocardial infarction, unspecified site*.

Using standardized ICD-9 codes allows hospitals to:

- Record diagnoses consistently
- Reduce ambiguity in disease names
- Retrieve patients with similar medical conditions
- Compare disease information across hospitals

- Produce morbidity and mortality statistics
- Support medical research and public-health reporting
- Support administrative and billing activities

Although newer revisions such as ICD-10 and ICD-11 are now available, this project uses ICD-9 because it models a hospital dataset based on that classification system.

1.3 ICD-9 Dictionary

The system must maintain a central **ICD-9 Dictionary**. Each record in this dictionary contains:

- ICD-9 code
- Official diagnosis title
- Description
- Disease category

The ICD-9 code is the primary key of the ICD-9 Dictionary.

The Diagnosis entity does not repeat the complete ICD-9 description. Instead, it stores the ICD-9 code as a foreign key referencing the corresponding record in the ICD-9 Dictionary.

For example:

ICD-9 Code	Official Description
401.9	Unspecified essential hypertension
410.9	Acute myocardial infarction, unspecified site
250.00	Diabetes mellitus without complication
272.4	Other and unspecified hyperlipidemia

Many diagnosis records may reference the same ICD-9 code. For example, many patients may receive the code 401.9, but the official description of that code is stored only once in the ICD-9 Dictionary.

1.4 Detailed Patient-Care Scenario

The following scenario illustrates how the Hospital Management System supports a patient's complete journey through the hospital.

Step 1: Patient Arrival and Identification

Suppose that a patient named **Daniel Tremblay** arrives at the hospital complaining of severe chest pain, shortness of breath, and dizziness.

At the hospital entrance, a registration employee or nurse searches for Daniel's existing patient record using information such as:

- Health insurance number
- Patient identifier
- Full name
- Date of birth
- Telephone number

If Daniel has previously received services from the hospital, the system retrieves his existing patient record. If he is visiting the hospital for the first time, a new patient record and a unique patient identifier are created.

The patient identifier is permanent and connects all of Daniel's past and future admissions, diagnoses, examinations, treatments, clinical notes, and discharge records.

Step 2: Creation of an Admission

After identifying the patient, the nurse creates a new **Admission**. The admission represents Daniel's current hospital visit and receives a unique admission identifier.

The nurse records information such as:

- Admission date and time
- Reason for the visit
- Admission type
- Admission source
- Insurance information
- Patient's current condition
- Name of the admitting healthcare professional

The admission type may be classified as emergency, urgent, elective, outpatient, newborn, or another type defined by the hospital.

Step 3: Nurse Triage and Initial Assessment

A triage nurse performs an initial assessment to determine the seriousness of Daniel's condition and establish his priority for treatment.

The nurse records:

- Blood pressure
- Heart rate
- Respiratory rate
- Body temperature
- Oxygen saturation
- Pain level
- Reported symptoms
- Known allergies
- Current medications

Based on the assessment, the nurse assigns a **triage priority level**. Because severe chest pain may indicate a life-threatening cardiac condition, Daniel is classified as a high-priority emergency patient.

Step 4: Selection of the Care Pathway

The patient's care pathway depends on the result of the triage assessment.

- A patient with a critical or potentially life-threatening condition is transferred to the **Emergency Department**.
- A patient who requires hospital care but is not in critical condition may be transferred to a **general medical ward**.
- A patient who requires only consultation or minor treatment may be examined, treated, and discharged without receiving a hospital bed.
- A patient requiring specialized examinations may be sent for laboratory testing, radiology examinations, or other diagnostic procedures.

Because Daniel's condition is potentially life-threatening, he is immediately transferred to the Emergency Department.

Step 5: Emergency Department Assessment

In the Emergency Department, a physician examines the patient and reviews his symptoms, vital signs, allergies, current medications, and previous medical history.

The physician creates a **Clinical Note** describing:

- The patient's symptoms
- The physical examination
- Initial clinical findings
- The suspected medical condition
- Planned diagnostic examinations
- The initial treatment plan

Based on the patient's symptoms, the physician writes "Diagnostic Orders". The physician request examination such as an electrocardiogram, blood tests, a chest X-ray, and a CT examination.

Step 6: Diagnostic Orders

Each requested examination is recorded as a **Diagnostic Order**. A diagnostic order records:

- Patient identifier
- Admission identifier
- Requesting physician
- Requested examination
- Request date and time
- Clinical reason for the examination
- Priority
- Order status

One admission may contain several diagnostic orders, while each diagnostic order belongs to exactly one admission.

Step 7: Radiology Examination

Daniel is transferred to the Radiology Department for a chest X-ray and CT examination.

For each examination, the system creates a **Radiology Examination** record containing:

- Examination identifier
- Examination type
- Examination date and time
- Body area examined
- Referring physician
- Radiology technician
- Examination status
- Link to the resulting medical images

After reviewing the images, a radiologist creates a **Radiology Report**. The report contains the radiologist's findings, interpretation, and clinical conclusion.

Step 8: Preliminary Diagnosis and Treatment Decision

The emergency physician reviews the electrocardiogram, blood-test results, chest X-ray, and CT examination.

The results suggest that Daniel has experienced an acute myocardial infarction. The physician creates a preliminary diagnosis linked to Daniel's current admission.

The diagnosis record includes:

- Diagnosis identifier
- Patient identifier
- Admission identifier
- ICD-9 code
- Diagnosis date and time
- Diagnosis type
- Diagnosing physician

For this condition, the physician may assign ICD-9 code 410.9, representing an acute myocardial infarction with an unspecified site.

Because Daniel requires continuous cardiac monitoring, the physician transfers him to the **Coronary Care Unit (CCU)**.

Step 9: CCU or ICU Stay

A new **Unit Stay** is created when Daniel enters the CCU. It records:

- Unit-stay identifier
- Admission identifier
- Unit type
- Ward and room
- Bed number
- Entry date and time
- Exit date and time
- Responsible physician
- Reason for transfer

While Daniel remains in the CCU, nurses record his vital signs, medications, observations, and responses to treatment. Physicians create progress notes and may request additional diagnostic examinations.

If Daniel had developed severe respiratory failure or another condition requiring advanced life support, he might have been transferred to the **Intensive Care Unit (ICU)**.

If surgery were necessary, a **Surgical Procedure** record would also be created.

Step 10: Transfer to a General Ward

After two days of monitoring and treatment, Daniel's condition improves. The physician determines that continuous cardiac monitoring is no longer required.

Daniel is transferred from the CCU to a general medical ward. The system records his CCU exit time and creates a new unit-stay record for the general ward.

In the general ward:

- Nurses continue monitoring the patient.
- Physicians perform follow-up examinations.
- Medications and treatments are administered.
- Progress notes are created.
- Additional tests may be requested.
- The patient's recovery is evaluated.

Step 11: Assignment of Final ICD-9-Coded Diagnoses

Before discharge, the physician reviews Daniel's complete medical record and confirms his final diagnoses.

Every completed admission must have at least one **primary diagnosis**. This diagnosis represents the main condition responsible for the patient's admission. The patient may also have zero or more **secondary diagnoses**, representing additional conditions identified or treated during the admission.

For example, Daniel's diagnosis records may include:

- ICD-9 code 410.9: Acute myocardial infarction, unspecified site — primary diagnosis.
- ICD-9 code 401.9: Unspecified essential hypertension — secondary diagnosis.
- ICD-9 code 272.4: Other and unspecified hyperlipidemia — secondary diagnosis.

Each diagnosis record connects:

Patient → Admission → Diagnosis → ICD-9 Code

This design is important because an ICD-9 code describes a medical condition, not the identity of the patient. The same patient may have multiple diagnoses, and the same ICD-9 code may be assigned to many different patients.

Step 12: Discharge

When Daniel is medically stable, the responsible physician authorizes his discharge.

A **Discharge Record** is created and includes:

- Discharge date and time
- Discharge destination
- Patient's condition at discharge
- Primary ICD-9 diagnosis
- Secondary ICD-9 diagnoses
- Prescribed medications
- Follow-up instructions
- Scheduled follow-up appointments
- Responsible physician

The physician also creates a discharge summary describing the reason for admission, important examination results, treatments provided, unit transfers, ICD-9-coded diagnoses, and recommendations for future care.

The admission status is changed to **discharged**. All information remains connected to Daniel's permanent patient record and can be reviewed during future hospital visits.

Summary of Relationships

- A **Patient** may have zero or more **Admissions**, while each admission belongs to exactly one patient.
- Each **Admission** represents one hospital encounter and has a unique admission identifier.
- An **Admission** may have one or more **Triage Assessments**.
- An **Admission** may generate zero or more **Clinical Notes**.
- An **Admission** may contain zero or more **Diagnostic Orders**.
- A **Diagnostic Order** may produce zero or more **Diagnostic Examinations**.
- An **Admission** may include zero or more **Radiology Examinations** and radiology reports.
- An **Admission** may contain zero or more **Surgical Procedures**.
- An **Admission** may include zero or more **Unit Stays**, such as Emergency Department, ICU, CCU, or general-ward stays.
- A completed **Admission** must have at least one **Diagnosis**, identified as the primary diagnosis.
- An **Admission** may also have zero or more secondary diagnoses.
- Every **Diagnosis** belongs to exactly one patient and one admission.
- Every **Diagnosis** references exactly one code in the **ICD-9 Dictionary**.
- One **ICD-9 Code** may be referenced by diagnoses belonging to many patients and many admissions.
- A **Patient** may therefore be associated with multiple ICD-9 codes through their diagnosis records.
- A completed **Admission** may have one **Discharge Record**, while each discharge record belongs to exactly one admission.

Together, these relationships model the complete patient-care lifecycle, including patient identification, admission, triage, emergency or general care, diagnostic examinations, radiology, surgery, ICU or CCU monitoring, ward transfers, ICD-9-coded diagnoses, clinical documentation, and discharge.

2 Generative AI Use Policy

Using AI and Generative AI tools, such as ChatGPT, Claude, Gemini, etc., can enhance the learning experience for students. However, any use of such tools for phase 1 of the project is only allowed for Section “Data sources”. Use of these tools in the other sections will be strictly prohibited and will constitute a violation of the Academic Code of Conduct.

3 Data sources

You are allowed to use generative AI tools to create your dataset. When prompting a generative AI tool, you must clearly instruct it to generate **completely fictional and synthetic hospital data created only for educational and learning purposes**. The generated data must not represent real patients or contain personally identifiable information.

You may ask a generative AI tool to generate data for any table required by your database, including:

- Patients
- Admissions
- Triage assessments
- Healthcare professionals
- Hospital units and unit stays
- Diagnostic orders
- Radiology examinations
- Surgical procedures
- Clinical notes
- Discharge records
- Diagnoses and ICD-9 codes

The generated data must be realistic enough to demonstrate the hospital scenario, but it must remain entirely fictional. You are responsible for reviewing the generated data and ensuring that:

- Primary-key values are unique.
- Foreign-key values reference existing records.
- Dates and timestamps follow a logical chronological order.
- A discharge time occurs after the corresponding admission time.

- A unit exit time occurs after its unit entry time.
- Procedures and examinations occur during the corresponding admission.
- Every completed admission has at least one primary ICD-9-coded diagnosis.
- The data satisfies all constraints defined in the database schema.
- The dataset contains enough variation to produce meaningful results for all required SQL queries.

You may also use the **MIMIC-III dataset** as an optional supporting data source, **but it is not mandatory to use this dataset**. You may use selected MIMIC-III tables or records to understand the structure of realistic hospital data, supplement your synthetic dataset, or populate relevant parts of your database. However, you are not required to reproduce the complete MIMIC-III schema. MIMIC-III dataset documents are available at the following link: [MIMIC-III Documentation](#).

The MIMIC-III supporting dataset is available at: [MIMIC-III Data](#)

4 Project phases

This project consists of two phases:

- Phase I addresses the relational database. In this phase, you will design, implement, and populate the relational database from the mentioned scenario.
- Note that the data of Phase I will be used to for Phase II. While you submit Phase I of your project now, during Phase II, you may enhance and refine your database to address unforeseen design issues. The final presentation will cover both phases.

4.1 Task 1: Drawing ER Diagram

For this task, you are required to carefully read the project overview description, and construct an Entity-Relationship (ER) diagram. Your diagram must include all relevant entities, their attributes, and the relationships among them. Be sure to clearly specify the primary keys and, where applicable, foreign keys within your diagram. The ER diagram must be created using a digital tool of your choice (e.g., Lucidchart, Draw.io, Microsoft Visio, or any other ER modeling software). Handmade or scanned diagrams will not be accepted.

4.2 Task 2: Making the Database

Based on the ER diagram created in the previous section, you are required to design and implement the hospital database. All entities, attributes, and relationships represented in your ER diagram must be properly mapped into relational tables. Your database schema must include appropriate primary keys, foreign keys, data types, uniqueness constraints, nullability constraints, and other integrity constraints required by your design.

After creating the database schema, you must populate all tables with sufficient data to test the database and execute the required SQL queries.

Whether you use AI-generated synthetic data, selected MIMIC-III data, or a combination of both, the imported data must be transformed as necessary to match your own database schema. You must ensure that the final data is correctly loaded, internally consistent, and suitable for executing all required SQL queries.

You must also briefly document:

- The source of the data used for each table
- Whether the data was synthetically generated or obtained from MIMIC-III
- Any prompts used to generate synthetic data
- Any cleaning or transformation performed before importing the data
- The method used to import the data into the database

You may use any database management system (DBMS) of your choice, such as MySQL, Oracle Database, PostgreSQL, or another suitable system.

4.3 Task 3: Writing SQL Queries

Using the Hospital Management System database developed in the previous tasks, write SQL queries to answer the following questions.

For each question, execute the query on your database and provide:

- The SQL query
 - A short comment explaining the query logic
 - The resulting output
1. List the names and admission dates of all patients who were classified as **high-priority emergency patients** during triage.
 2. Find all patients who had **more than three hospital admissions**. Display the patient identifier, patient name, and total number of admissions.
 3. Retrieve the names, admission dates, and discharge dates of patients who were discharged without undergoing any **surgical procedure** during the admission.
 4. List all patients who had both a **radiology examination** and a **surgical procedure** during the same admission.
 5. Find all ICU or CCU unit stays that lasted **more than seven days**. Display the patient name, unit type, unit entry time, unit exit time, and calculated length of stay.
 6. Count the total number of hospital admissions for each patient. Include patients who have no recorded admissions.

7. List all patients who entered the hospital through the **Emergency Department** and were later transferred to an **ICU or CCU** during the same admission.
8. Retrieve the **most frequently assigned primary ICD-9 diagnosis** among admissions that included at least one ICU or CCU stay. Display the ICD-9 code, its official description, and the number of times it was assigned.
9. Calculate the **average length of stay** for each critical care unit type, such as MICU, SICU, ICU, and CCU. Display the unit type and its average length of stay.
10. List all patients who underwent a **surgical procedure before entering an ICU or CCU** during the same admission. Display the patient name, procedure date and time, critical care unit type, and unit entry time.
11. Retrieve the name of each patient and the **total number of radiology examinations** received across all admissions. Include patients who have not received any radiology examinations and display zero for their examination count.
12. Find all patients whose **discharge-summary clinical note** contains the keyword “recovery”. The search should be case-insensitive.
13. List all admissions in which the patient had at least one **radiology examination** but had no recorded **ICU or CCU stay**.
14. Retrieve the patient or patients with the **longest completed hospital stay**, calculated from admission time to discharge time. If multiple patients have the same maximum length of stay, display all of them.
15. Count the total number of **transfers between hospital units** for each patient. The patient’s first unit placement is not considered a transfer; each subsequent unit stay within the same admission represents one transfer.
16. List all patients who stayed in **more than one type of critical care unit** during the same admission. For example, a patient may have been transferred from an ICU to a CCU.
17. Find all admissions during which the patient received **more than one distinct ICD-9 diagnosis**. Display the patient name, admission identifier, and number of distinct ICD-9 codes.
18. Retrieve the **latest clinical note for each patient**. Display the patient name, note type, note creation date and time, and note contents.
19. List all admissions where the patient’s discharge status or discharge condition indicates that the patient **died during the hospital stay**. Display the patient name, admission date, recorded date of death, and associated primary ICD-9 diagnosis.

20. Find all patients who underwent a **surgical procedure and a radiology examination on the same calendar day** during the same admission. Display the patient name, admission identifier, procedure type, radiology examination type, and the common date.

Instructions

- Write **one SQL query** for each question.
- Add comments to each SQL query to explain its main logic.
- Ensure that the SQL syntax is correct, properly indented, and readable.
- Data cleaning is not necessary for this phase.
- Use appropriate SQL operations, including JOIN, LEFT JOIN, GROUP BY, HAVING, subqueries, aggregate functions, date and time functions, and common table expressions where necessary.
- When a query asks for patients with no related records, use an appropriate technique such as LEFT JOIN, NOT EXISTS, or a subquery.
- When calculating a length of stay, use the difference between the appropriate entry and exit timestamps.
- Only completed admissions with a recorded discharge time should be considered when calculating the total hospital length of stay.
- Each completed admission must have at least one primary ICD-9-coded diagnosis. A patient may have multiple diagnoses during the same admission.
- All 20 queries must be completed. Depending on the team size, each student must be responsible for implementing approximately **four to five queries**.
- The contribution of each team member must be clearly identified in the submitted report.
- Every team member must understand all submitted queries, even if the queries were initially divided among different members.

5 Submission Guidelines

1. Submit a PDF file, a well-structured document that follows the sections defined in this handout and uses a professional academic writing style. The Documentation including a clear drawing of the ER diagram. SQL queries include code and a screenshot of the result for Task 3. In this document, please ensure that you paste your code for each task into the specified query (from Q1 to Q20). Ensure that your file is named according to the following format: `SOEN363.Project.Phase1.YourTeamName.zip`.

2. Submit your assignment via Moodle before the due date.
3. The project must be demonstrated online to the marker at the specified date and time. This schedule will take place within one week after the due date, and further details regarding the specific date and time will be announced later. Make sure to keep an eye out for those updates!

6 Rubric

Table 1: Rubric.

Task	Description	Points
	Drawing ER diagram	20 points
Task 1	All entities, attributes, and relationships included, PKs and FKs clearly defined, and correct cardinality (10pt). Clean, well-labelled diagram made with a digital tool (10pt).	
	Creating database and importing the data	10 points
Task 2	Schema fully matches ER diagram. Tables, PKs, FKs, constraints correct. Data imported accurately and completely. No integrity issues.(10pt).	
	SQL Queries	40 points
Task 3	Each query 2 pt .	
	Documentation & Presentation	30 points
Task 4	- A PDF file, a well-structured document that follows the sections defined in this handout and uses a professional academic writing style. The Documentation including a clear drawing of the ER diagram. SQL queries include code & output. Professional formatting. (10pt). - Presentation and attendance to the demo (20pt).	
Total		100 points